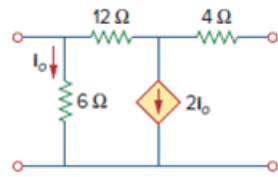


Problem

Determine the y parameters for the circuit in Fig. 19.19.

Figure 19.19



Step-by-step solution

Step 1 of 10

Refer to Figure 19.19 in the text book.

Connect the current source $\mathbf{I}_1$ as shown in Figure 1. Identify the currents and voltages in the circuit.

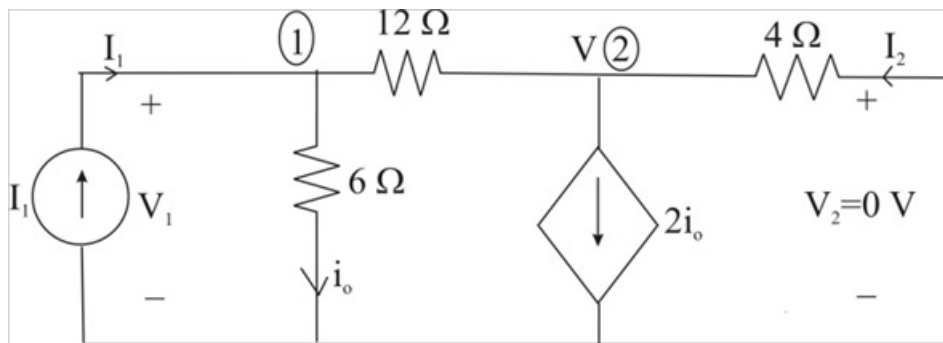


Figure 1

Step 2 of 10

Consider the nodal equation at node 2.

$$\frac{V}{4} + 2i_o + \frac{V - V_1}{12} = 0$$

Substitute $\frac{V_1}{6}$ for i_o .

$$\frac{V}{4} + 2\left(\frac{V_1}{6}\right) + \frac{V - V_1}{12} = 0$$

$$\frac{V}{4} + \frac{V_1}{3} + \frac{V - V_1}{12} = 0$$

$$3V + 4V_1 + V - V_1 = 0$$

$$4V + 3V_1 = 0$$

$$V = \left(\frac{-3}{4}\right)V_1$$

Therefore, the expression for the voltage $\mathbf{V}$ is $\left(\frac{-3}{4}\right)V_1$.

Step 3 of 10

Consider the nodal equation at node 1.

$$\frac{V_1}{6} - I_1 + \frac{V_1 - V}{12} = 0$$

$$2V_1 - 12I_1 + V_1 - V = 0$$

$$3V_1 - 12I_1 - V = 0$$

Substitute $\left(\frac{-3}{4}\right)V_1$ for V .

$$3V_1 - 12I_1 - \left(\frac{-3}{4}\right)V_1 = 0$$

$$V_1 - 4I_1 - (-0.25)V_1 = 0$$

$$V_1 + (0.25)V_1 = 4I_1$$

$$(1.25)V_1 = 4I_1$$

$$V_1 = 3.2I_1$$

Therefore, the expression for the voltage V_1 is $I_1(3.2)$.

Step 4 of 10

Consider the expression for the current I_2 .

$$I_2 = -\frac{V}{4}$$

Substitute $\left(\frac{-3}{4}\right)V_1$ for V .

$$I_2 = -\frac{1}{4}\left(\frac{-3}{4}\right)V_1$$

$$= \left(\frac{3}{16}\right)V_1$$

Therefore, the expression for the current I_2 is $\left(\frac{3}{16}\right)V_1$.

Step 5 of 10

Determine the expression for the short circuit input admittance y_{11} .

$$y_{11} = \frac{I_1}{V_1}$$

Substitute $I_1(3.2)$ for V_1 .

$$Y_{11} = \frac{I_1}{I_1(3.2)}$$

$$= 312.5 \text{ mS}$$

Therefore, the expression for the short circuit input admittance y_{11} is 312.5 mS.

Step 6 of 10

Determine the value of the short circuit transfer admittance y_{21} from port-2 to port-1.

$$y_{21} = \frac{I_2}{V_1}$$

Substitute $\left(\frac{3}{16}\right)V_1$ for I_2

$$y_{21} = \frac{\left(\frac{3}{16}\right)V_1}{V_1}$$

$$= \frac{3}{16}$$

$$= 187.5 \text{ mS}$$

Therefore, the value of the short circuit transfer admittance y_{21} from port-2 to port-1 is

$$\boxed{187.5 \text{ mS}}$$

Step 7 of 10

Connect the current source I_2 as shown in Figure 2. Identify the currents and voltages in the circuit.

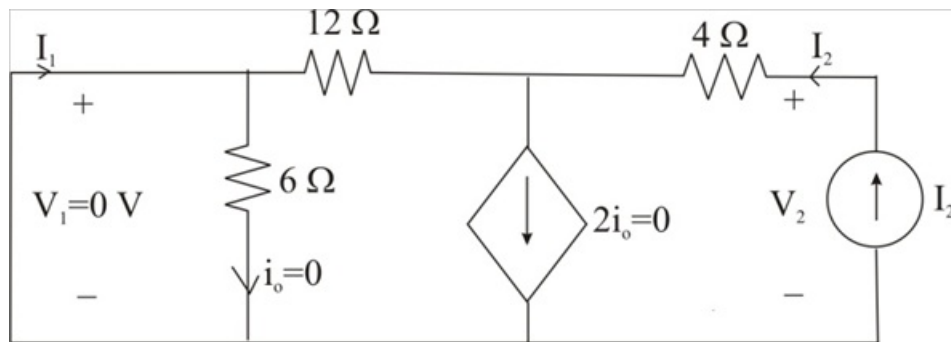


Figure 2

Step 8 of 10

Notice that the current i_o is zero since 6Ω resistor is shorted. Hence, the value of the dependent source is zero that is, open circuited.

Consider the expression for the voltage V_2 .

$$\begin{aligned} V_2 &= I_2(4+12) \\ &= I_2(16) \end{aligned}$$

Therefore, the expression for the voltage V_2 is $I_2(16)$.

Consider the expression for the current I_1 .

$$I_1 = -I_2$$

Therefore, the expression for the current I_1 is $-I_2$.

Step 9 of 10

Determine the value of the short circuit transfer admittance y_{12} from port-1 to port-2.

$$y_{12} = \frac{I_1}{V_2}$$

Substitute $I_2(16)$ for V_2 and $-I_2$ for I_1 .

$$\begin{aligned} y_{12} &= \frac{-I_2}{I_2(16)} \\ &= -62.5 \text{ mS} \end{aligned}$$

Therefore, the value of the short circuit transfer admittance y_{12} from port-1 to port-2 is

$$\boxed{-62.5 \text{ mS}}.$$

Step 10 of 10

Determine the expression for the short circuit output admittance y_{22} .

$$y_{22} = \frac{I_2}{V_2}$$

Substitute $I_2(16)$ for V_2 .

$$\begin{aligned} y_{22} &= \frac{I_2}{I_2(16)} \\ &= 62.5 \text{ mS} \end{aligned}$$

Therefore, the expression for the short circuit output admittance y_{22} is $\boxed{62.5 \text{ mS}}$.